



ed IT support to a much more complete coverage of communication and information-retrieval activities.

Parametric design

Parametric design is a method wherein computers are used to intelligently design architectural objects based on associations and rules. First, definitions are set in parametric software and then simply manipulated to rapidly generate multiple iterations of the design in 3D.

Aerospace industries had been applying computers to calculate complex, warped surfaces and animated flight-path simulations, and this attracted architects. Antonio Gaudi, Erich Mendelsohn, Frei Otto, Kiesler and Kiyonori Kikutake were some great artists and architects who had imagined and modelled complex structures and forms with different degrees of technical proficiency.

In 1960s, Roig recognised that new computer technologies could assist their design and construction. However, it was only by 1980s that breakthroughs in parametric design became useful to architects. Advances in the quasi-scientific field of plant and animal morphology supported novelty that could be applied with imagination to tectonic practices.

Harbin Opera House, located in the northern Chinese city of Harbin, and Walt Disney Concert Hall by Frank O. Gehry and Partners in Los Angeles, USA (one of the most acoustically-sophisticated concert halls in the world) are examples of buildings that use parametric design, partly or wholly.

Use of this tool allows for more complex free-form shapes as well as multiple reactive yet repeating elements to be fashioned. Parametric design has provided architects a very useful tool that allows them to design and construct innovative buildings with more exacting qualitative and quantitative conditions.

Constructing green buildings using IT

Global warming is being caused by an increase in average temperature of Earth's surface; this average temperature rose about 0.9°C during the last century due to greenhouse effect. Scientific studies indicate average temperature would rise a further 1.1°C to 4.5°C during the 21st century depending upon the increase in GHG emissions (mostly consisting of CO_2).

optimum operating conditions. It is also behind Freedom Tower, also known as One World Trade Center, New York, USA, in terms of making it an energy-efficient building. The tower's design features two complex pleated-skin facades. The office space section of the tall and twisting building has been modelled using ArchiCAD to import the intricate building skin geometry into an energy-analysis application, EnergyPlus. Performances of different building skin options are simu-

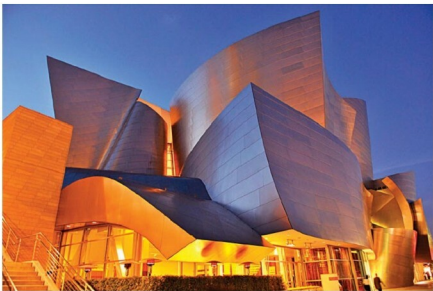


Fig. 2: Walt Disney Concert Hall, Los Angeles, USA (Image courtesy: www.homedit.com)

Every unit of electricity (kWh) used generates 0.82kg of CO_2 at the power station. One of the most challenging issues architects face today is controlling energy consumption in buildings. In developed countries such as the USA and the UK, IT is extensively being used to create intelligent and energy-efficient building designs. There are computer tools that enable architects to precisely estimate the energy consumption in a given building. IT also provides evaluation of practically infinite options and helps in shortlisting the most efficient one.

Additionally, IT can also be used to control the operation of energy-intensive devices and services for

lated, as an integral part of the rest of the building, while EnergyPlus determines the optimal day lighting design and the overall building energy performance.

In India too, more and more green buildings are being constructed using IT applications. Computer technology is very useful for architects, as structures impossible to draw using pen and paper can be conceived and detailed with the help of technology.

A lot of software have been developed to simulate a building in various conditions of external weather, internal air cooling/heating and energy loads of different activities being conducted inside to